

Properties of Sinusoidal Graphs

Trigonometric functions can model periodic phenomena.

EXAMPLE

The tides on the French island of Mont Saint-Michel in the Couesnon River can be approximated by a sinusoidal function.

- A** Write an equation modeling the height of the tides if the high tide of 12.5 m occurs every 12 h and it is now low tide when the river is 0.5 m below sea level.

SOLUTION

Let y = water level (meters) and x = time (hours).

$$\text{equation of midline: } y = \frac{12.5 + (-0.5)}{2} = 6$$

So $C = 6$.

$$\text{amplitude} = \frac{12.5 - (-0.5)}{2} = 6.5$$

The tide returns to the maximum after 12 h, so the period is 12.

To find B , solve for B in the period formula.

$$\text{period} = \frac{2\pi}{B}$$

$$12 = \frac{2\pi}{B}$$

$$12B = 2\pi$$

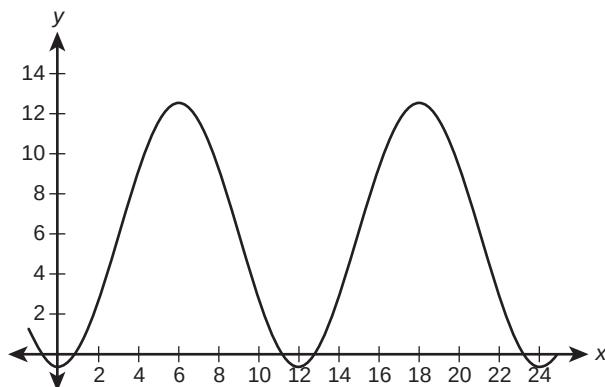
$$B = \frac{\pi}{6}$$

Since the function starts at a minimum, a cosine function where $A < 0$ works best, so $A = -6.5$. Thus, the equation that models the tides is

$$y = -6.5 \sin\left(\frac{\pi x}{6}\right) + 6.$$

B Sketch the graph of the function for the first 24 h.

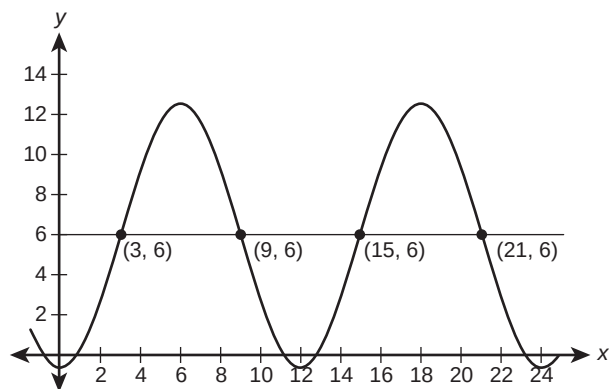
SOLUTION



C If the daily mail delivery can only be made by boat when the water is above 6 m, between what times can the mail delivery be made if the initial low tide occurs at midnight?

SOLUTION

Since the midline is $y = 6$, you must find the intersections of the curve with the midline. Using the graph, the intersections occur at $x = 3, 9, 15$, and 21 during the first 24 h.



The curve is above the midline on the intervals $(3, 9)$ and $(15, 21)$, so the boat can make the delivery to the island between 3 a.m. and 9 a.m., and between 3 p.m. and 9 p.m. ■